

SONIX: Real-Time Detection of Voice-Cloning Impersonation in Indian-Language Calls

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SIH26104

Abstract. Voice-cloning tools now produce convincing speech from a few seconds of a person’s audio, and impersonation calls use it against banks, contact centres and families. SONIX is a detection layer that scores live call audio every 0.5 s and raises a graded alert for a human operator without ever blocking the call. It runs as a service with a consent gate, a silence gate, an audit trail and two independent clients: an operator console and a browser extension for web calls. While building the detector we found that heads trained on a frozen wav2vec2 XLS-R embedding learn the wrong signal. The mean-pooled embedding identifies the spoken language with 99.33% accuracy across 12 languages and the source corpus of genuine speech with 100% accuracy, while an English-trained spoof classifier falls from AUC 0.9992 in-domain to 0.5196 on Indian speech; removing 20 language directions leaves it at 0.5011. On matched genuine and cloned recordings of the same speakers, the heads trained on the most Indian speech ranked worst. We rebuilt the detector around these measurements: mean and standard-deviation pooling over three layers, genuine and synthetic speech in every training language, a loss balanced over language, generator and label with worst-group model selection, and channel conditioning (telephone codecs, room acoustics, noise, RawBoost) applied identically to both classes. The build also exposed two data artefacts that inflate results, silence padding and speaker overlap between standard splits, and this report describes the evaluation protocol that controls for them. The report closes with SONIX v4, the final model built from these findings, and the full record of how it was assembled.

Keywords: Anti-spoofing · Voice cloning · Deepfake speech · Self-supervised speech models · Indian languages · Shortcut learning · Real-time systems

1 Introduction

In 2024, Indian citizens reported losses of INR 22,845 crore to cyber fraud across 36.4 lakh incidents, three times the INR 7,465 crore reported for 2023 [1]. Voice impersonation is a growing part of that surface. In a 2023 survey of 1,010 Indian adults, 47% had faced an AI voice scam or knew someone who had, 69% were not confident they could tell a cloned voice from a real one, and 83% of those targeted lost money; the same study produced an 85% voice match from three seconds of audio [2]. Contact-centre operators see the same trend: deepfake call attempts rose by more than 1,300% during 2024 across 1.2 billion analysed calls [3].

A caller cannot hear the difference, and nothing on the line checks for it. Caller ID can be spoofed, recognising a familiar voice is exactly what the attack exploits, and fraud monitoring only sees the transaction after the call. Problem Statement SIH26104 asks for a detector that works *during* the call, handles Indian languages and accents, and is exposed through APIs so that banks, contact centres and telecom systems can use it as a shared security layer.

Automatic spoofing detectors do exist, but most are developed and scored on English benchmarks such as ASVspoof 2019 LA [5]. Trained on ASVspoof, even strong systems degrade by an order of magnitude on real-world recordings [8], and a recent survey reports open-set error rates of 4.54–13.48% EER for the best current systems [4]. For an Indian deployment the more important question is *what* a detector has learned. When we trained heads on a frozen XLS-R embedding [9], in-domain scores were near perfect while matched Indian recordings scored close to chance, and the heads that had seen the most Indian speech did worst.

This report documents the whole project: the real-time system, the development history of the detector, the diagnosis of why it failed, the redesigned model, the data engineering behind it and the evaluation protocol. Our contributions are:

1. A working real-time service that scores live audio every 0.5 s, with a server-enforced consent gate, a silence gate, an audit trail without stored audio, swappable detector heads and two independent clients (Sect. 4).
2. A controlled diagnosis showing that the standard mean-pooled SSL embedding encodes language and corpus almost perfectly while carrying no synthesis signal that transfers across languages (Sect. 6).
3. A detector redesigned around that diagnosis, covering representation, data composition, objective, model selection and channel conditioning (Sect. 7).
4. A data audit that found and removed two artefacts, silence padding and speaker overlap, that would otherwise inflate reported results (Sect. 8).
5. SONIX v4, the final model, with a complete record of its data, training configuration and acceptance checks (Sect. 9), and an evaluation protocol with speaker-disjoint Indian test data and matched genuine and cloned recordings, applied to every earlier SONIX head (Sects. 10 and 11).

2 Background and Related Work

2.1 Benchmarks and the Generalisation Gap

The ASVspoof challenges define the standard protocol for spoofing countermeasures. ASVspoof 2019 LA [5] contains English bona fide speech from VCTK speakers and spoofs from 19 text-to-speech and voice-conversion systems, with six attacks in training and mostly unseen attacks in evaluation. ASVspoof 2021 [6] added transmission channels and a deepfake track with compressed audio, and ASVspoof 5 [7] extends this further. Müller et al. [8] collected the In-the-Wild dataset: 37.9 hours of genuine and deepfaked speech from 58 public figures. Models that reached 1.2–3.2% EER on ASVspoof 2019 LA scored around 37% EER on it, a degradation of up to one thousand percent. Sun et al. [4] identify this domain shift as the dominant failure mode and warn

about “benchmark overfitting”, where models exploit dataset quirks that do not carry over.

2.2 Self-Supervised Front-Ends

Large self-supervised speech models such as wav2vec 2.0 and its cross-lingual version XLS-R [9], pre-trained on 436 k hours in 128 languages, give strong spoofing features. Tak et al. [10] fine-tuned wav2vec 2.0 with an AASIST back-end [11] and RawBoost augmentation [14] and reported the lowest EERs at the time on ASVspoof 2021 LA and DF. Zhang et al. [13] weighted multiple XLS-R layers with a sensitive layer selection (SLS) classifier and reported 1.92% EER on 2021 DF, 2.87% on 2021 LA and 7.46% on In-the-Wild. Both results show that intermediate layers carry useful spoofing information, which our layer sweep (Sect. 6.1) confirms for Indian data.

2.3 Shortcut Learning and Group Robustness

A classifier that can solve its training task through a spurious feature often will [15]. Group distributionally robust optimisation [16] addresses this by weighting training groups and selecting models on worst-group performance, which is the principle behind our objective. Iterative null-space projection (INLP) [17] removes linearly decodable attributes from a representation and is a standard tool for testing whether a protected attribute masks other information.

2.4 Proactive and Passive Detection

AudioSeal [19] embeds an imperceptible watermark at generation time and localises it at sample level, but it only covers generators that insert the mark; its paper also reports a passive baseline at 0.704 accuracy and 0.194 false-positive rate on re-synthesised audio. SONIX is passive and makes no assumption about the generator. Bird and Lotfi [20] report real-time detection of converted voices, evaluated with a split in which segments of the same recording can appear in both training and test data; we avoid this by splitting at recording and speaker level.

2.5 Indian-Language Resources

IndicVoices [21] provides natural, read and conversational speech from thousands of speakers across Indian languages (CC BY 4.0). IndicSynth [22] provides about 4,000 hours of synthetic speech from 989 target speakers across 12 Indian languages. MLAAD [23] covers dozens of TTS models across many languages, including Indian ones (CC BY-NC 4.0). MMS [25] provides TTS for more than 1,000 languages. None of these, on its own, gives genuine and synthetic speech for the same languages from comparable channels, which is the gap our training set is built to close.

Table 1. Requirements of SIH26104 and how SONIX addresses them.

Requirement	SONIX design
Real-time, during the call	Streaming 4.0 s windows on a 0.5 s hop; smoothing and hysteresis; first score after one window
Indian languages and accents	Genuine and synthetic training speech in 8 Indian languages plus English; evaluation on unseen Indian speakers
APIs and SDKs	REST and WebSocket service with voice-over-IP, browser and upload inputs
Reusable security layer	One service used by independent clients; detector heads can be swapped without redeploying

3 Problem Setting and Design Principles

Problem Statement SIH26104 (AICTE Cyber Security Cell) asks for four properties, which we map to design decisions in Table 1.

Three principles follow from the setting. *Flag, never block.* A false alarm that cut off a genuine customer would be worse than no detector, so SONIX raises a Green, Amber or Red band and leaves the decision to a person. *Consent first.* No audio is scored until an operator approves the session, and the check is enforced on the server, not in the client. *Honest scope.* SONIX runs where call audio is legitimately available to software: contact-centre and VoIP calls through the API, and Google Meet or other web calls through the browser extension. Android and iOS do not let third-party apps read ordinary phone-call audio, so ordinary mobile calls are out of scope.

4 System Architecture

4.1 Capture Surfaces

The server (`realtime/server.py`, `aiohttp`) accepts audio in three modes. In *VoIP* mode it listens for Asterisk AudioSocket connections over TCP, so a PBX can fork a call leg to SONIX. In *WebRTC* mode it accepts 16-bit 16 kHz mono PCM over a WebSocket, which serves two browser clients: a microphone capture page and a Chrome extension. In *upload* mode it scores recorded files. The extension (Manifest V3, Chrome and Edge 116+) captures the audio of a Google Meet tab, which contains the other participants rather than the user’s own microphone. That is the right direction for this problem, since the question is whether the person on the other end is synthetic. The extension shows the band in its popup and as an overlay on the call page; all detection runs in the service.

4.2 Consent Gate and Session Lifecycle

Every call receives a six-digit pairing code that an operator must approve in the console. Until approval, the session discards every incoming audio chunk; codes expire after 120 s, and refusals are written to the audit trail. Only in the listening state is audio buffered and scored.

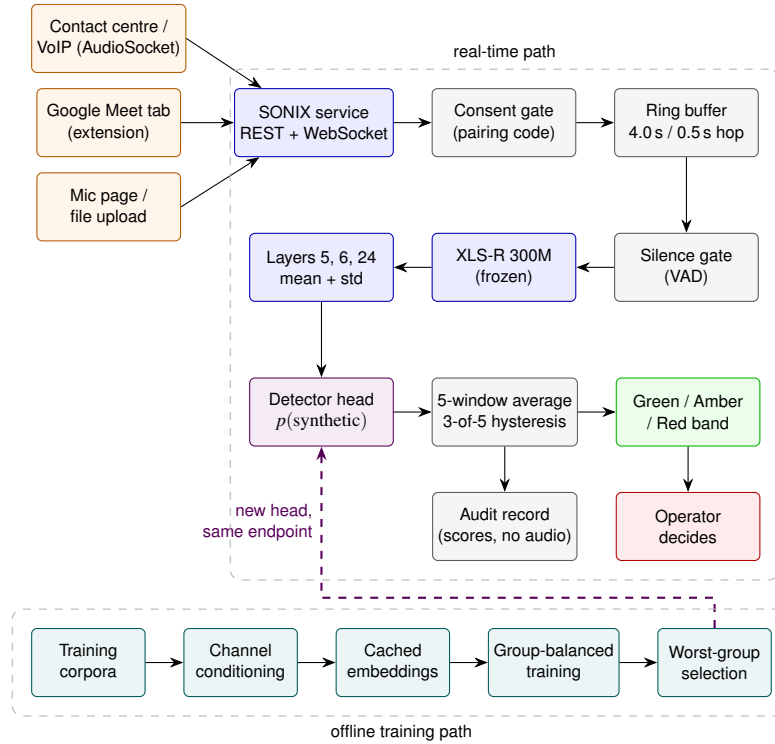


Fig. 1. SONIX architecture. Top: the real-time path from call audio to a graded alert for the operator. Bottom: the offline path that produces detector heads, which are loaded into the running service without retraining or redeploying the front-end. The band and scores are returned to the client that sent the audio.

4.3 Windowing and Silence Gate

A ring buffer of 64,000 samples holds the most recent 4.0 s at 16 kHz and emits a window every 0.5 s (87.5% overlap). A voice-activity detector splits each window into 25 ms frames, marks a frame as speech when its RMS energy clears a floor and its zero-crossing rate stays below a ceiling that separates voice from hiss, and passes the window only if enough frames are speech. Windows without speech are not scored, so silence cannot produce a verdict. Per-window RMS, peak and speech ratio are logged so the operator can see what the gate saw.

4.4 Scoring Engine and Heads

Windows are batched and scored off the event loop. The frozen XLS-R 300M front-end is loaded once and shared; each detector head is a small network with its own input standardiser, loaded from a checkpoint. The live path calls the same scoring implementation as the offline evaluation tools, so a live score and an offline score of

Table 2. Main service endpoints.

Endpoint	Purpose
/ws	WebSocket: binary PCM in, JSON scores and band out
/mic	Browser microphone capture page
/api/status	Server state, scoring availability, live telemetry
/api/telemetry	Per-call window and score history
/api/approve	Approve a pairing code (consent gate)
/api/end-call	End a call and write its audit record
/api/score-file	Score an uploaded recording

the same audio agree by construction. A registry names every trained head, and the console can score one clip with several heads side by side, which is how the comparison in Sect. 11 was run.

4.5 Smoothing and Risk Bands

Scores pass through a 5-window moving average and 3-of-5 hysteresis, so the band only changes when several consecutive windows agree. The service maps the smoothed score to Green below 0.35, Amber from 0.35 to 0.65 and Red from 0.65. These thresholds are provisional: they were not fitted on in-domain call audio, and the console exposes them as operator settings. Thresholds for the final model are fitted per domain on held-out data (Sect. 10).

4.6 Service Interface

Table 2 lists the main endpoints. Two clients use the service without sharing any code: a Streamlit operator console with a live risk timeline, telemetry, an audio-path view and pairing approval, and the Chrome extension.

4.7 Audit and Privacy

Each call writes a JSON record of its metadata, event trail and per-window scores. Audio is never written to disk: it lives only in the 4 s ring buffer and is overwritten continuously. Audio leaves the browser only to the SONIX service.

5 Development History

SONIX went through several detector heads before the final model. Each was a small MLP on the 1024-dimensional, mean-pooled final-layer XLS-R embedding: a linear layer to 256 units, ReLU, dropout 0.3 and a linear output (262,657 parameters), trained with class-weighted binary cross-entropy. They differ only in training data (Table 3).

Two observations from this phase drove the rest of the project. First, informal recordings made by the team on phones and laptops triggered false alarms on roughly

Table 3. Earlier SONIX detector heads. All use the same 1024-d representation and head architecture.

Head	Training data
<code>baseline</code>	ASVspoof 2019 LA train (clean)
<code>augmented</code>	+ G.711 μ -law copies of the training set
<code>robust</code>	+ RawBoost channel and impulsive-noise copies
<code>robust_v2</code>	+ room reverberation and MUSAN noise copies
<code>full_ho</code>	+ IndicVoices genuine speech (12% of Indian recordings held out)
<code>full_indic_as5</code>	+ IndicVoices + ASVspoof 5
<code>v3</code>	+ 35,200 Indian spoofs from three families (MMS-TTS, IndicSynth voice conversion, channel-augmented copies)

Table 4. Linear probes on the mean-pooled final-layer XLS-R embedding.

Probe	Controlled setting	Result	Chance
A: language	12-way; IndicSynth only, so every clip is synthetic and from one generator family; 12,600 train / 4,200 test	99.33%	8.33%
B: corpus	IndicVoices vs. ASVspoof; genuine clips only; split by recording; 11,125 train / 3,705 test	100.00%	82.54%
C: spoof, in-domain	trained and tested on English (ASVspoof), AUC	0.9992	0.50
C: spoof, transfer	trained on English, tested on Indian speech, AUC	0.5196	0.50

half of genuine clips with the `baseline` head, worst on clips with long quiet stretches. That motivated the silence gate and the channel augmentation in `robust` and `robust_v2`. Second, adding Indian speech did not behave as expected. `full_ho` almost never flagged genuine Indian speech, which looked like success, but it had only ever seen genuine Indian audio, so “Indian” had become a proxy for “genuine”. `v3` added Indian spoofs to fix that. Section 11 shows that on matched recordings both heads still ranked close to or below chance. The diagnosis in the next section explains why.

Two engineering failures from this phase are worth recording because both were silent. A blanket half-precision cast of the front-end produced embeddings that were entirely NaN while the extractor’s shape check reported success; extraction now checks every shard for non-finite values, and the trainer refuses NaN rows. The second, silence padding, is described in Sect. 8.

6 Diagnosis: What the Embedding Encodes

We trained linear probes (logistic regression) on the representation used by all earlier heads: the final-layer XLS-R output, mean-pooled over time. Each probe controls for a different confound (Table 4).

Probe A shows that language can be read off the vector almost perfectly even when nothing else varies. Probe B shows that genuine speech from the two corpora is perfectly separable: the label is constant, so the probe can only be using corpus properties such as channel, microphone and recording conditions. Probe C shows that an English-trained

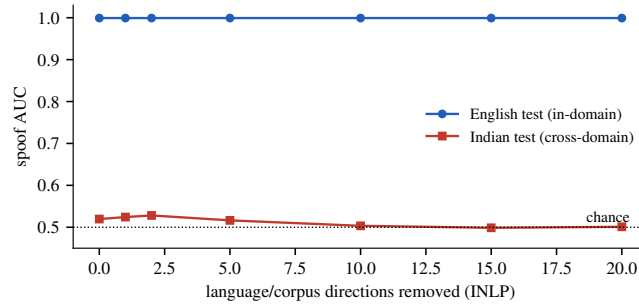


Fig. 2. Probe C after iterative null-space projection. Removing the directions that best predict language and corpus leaves in-domain AUC unchanged and moves cross-domain AUC towards chance.

spoof direction carries no ranking information on Indian speech. An AUC of 0.52 is not a well-ranked detector at a poor threshold; it is no ranking, so recalibration cannot help.

If language were masking a transferable spoofing signal, projecting it out would reveal that signal. Figure 2 shows the opposite. After 20 rounds of INLP [17], in-domain AUC stays at 0.9992 and cross-domain AUC ends at 0.5011. The in-domain score was never based on language; it was based on properties of ASVspoof itself, and none of them transfer.

Mechanism. XLS-R produces about 50 frames per second, so a 4 s window is roughly 200 frames averaged into one vector. Language, speaker, channel and noise level are stable across those frames and survive the average. The traces a vocoder leaves are short and local: discontinuities at frame boundaries, locally over-smoothed spectra, fine timing jitter. Averaging treats them as noise. The final layer compounds the problem, because it is the most content-abstracted layer in the stack.

6.1 Which Layers Keep the Signal

We repeated probe C on every hidden layer, pooled with mean and standard deviation, and measured AUC on ASVspoof attacks not seen in training (Fig. 3). Early-middle layers keep more of the signal: layer 6 reached 0.984, against 0.953 for the final-layer output used by the earlier heads. Every layer still decodes language and corpus at 100%, so layer choice alone is not a fix; it has to be combined with the data and objective changes below.

Caveats. Probe C trains on English only. It shows that the representation carries no synthesis signal that transfers across languages; it does not show that a head trained on Indian spoofs cannot detect *seen* Indian generators. That question is answered by the matched-recording test in Sect. 11.

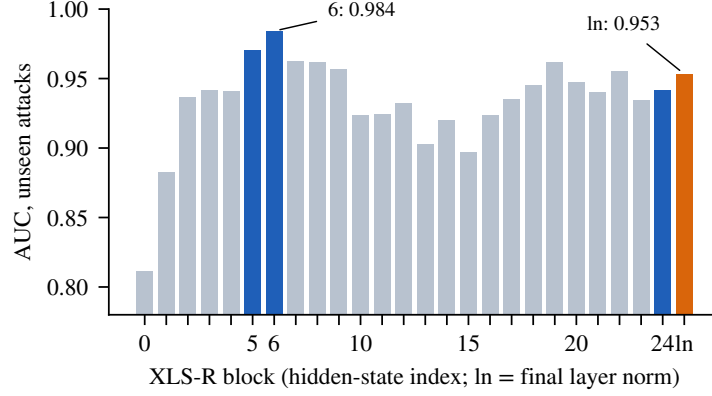


Fig. 3. Spoof-probe AUC on unseen ASVspoof attacks for each XLS-R block. Blue: blocks used by the final model (5, 6, 24). Orange: the final-layer output used by the earlier heads.

7 Method

7.1 Representation

For a 4 s window with hidden states $\mathbf{h}_{\ell,t}$ at layer ℓ and frame $t = 1, \dots, T$, we keep both the mean and the standard deviation over time,

$$\mu_{\ell} = \frac{1}{T} \sum_{t=1}^T \mathbf{h}_{\ell,t}, \quad \sigma_{\ell} = \left(\frac{1}{T} \sum_{t=1}^T (\mathbf{h}_{\ell,t} - \mu_{\ell})^2 \right)^{1/2}, \quad (1)$$

and concatenate $[\mu_5; \sigma_5; \mu_6; \sigma_6; \mu_{24}; \sigma_{24}]$ into a 6,144-dimensional vector. Layers 5 and 6 carried the most transferable signal in the sweep, and layer 24 keeps the higher-level view. The standard deviation keeps some of the frame-to-frame variation that the mean discards. The front-end stays frozen, so embeddings are extracted once and cached, and training a head takes minutes.

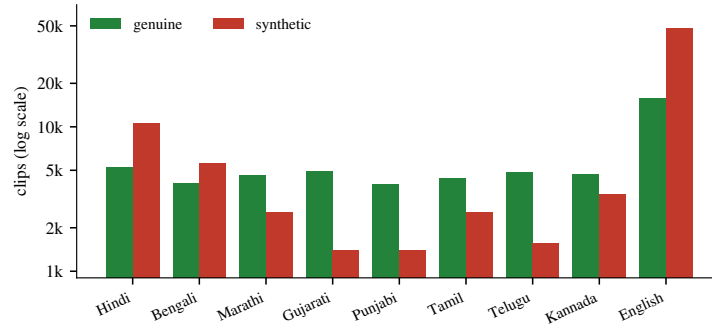
7.2 Training Data

The earlier heads saw genuine Indian speech without Indian spoofs (`full_ho`) or with Indian spoofs from few families (`v3`), and their English data was mostly ASVspoof. The final training set (Table 5) has genuine and synthetic speech in every language it contains, and spoofs from four distinct generator types: text-to-speech, voice conversion, neural vocoders and the ASVspoof attack systems.

Language coverage. Malayalam, Urdu, Odia and Sanskrit appear only as synthetic speech in the available corpora (IndicSynth and MLAAD), with no genuine speech in the same languages. Kept in training, their 7,600 clips would teach the head that these languages mean “fake”. We therefore drop them, which leaves 8 Indian languages plus English (Fig. 4). Adding genuine speech for these languages, for example from

Table 5. Training data before channel conditioning.

Corpus	Class	Content	Clips
IndicVoices [21]	genuine	8 Indian languages	36,859
LibriSeVoc [24]	genuine	English read speech	13,201
ASVspoof 2019 LA [5]	genuine	English	2,580
MLAAD [23]	synthetic	TTS, 19 models, Indian languages	19,000
IndicSynth [22]	synthetic	voice conversion, 12 languages	16,800
LibriSeVoc [24]	synthetic	6 neural vocoders, 4,200 each	25,200
ASVspoof 2019 LA [5]	synthetic	6 attack systems (A01–A06)	22,800
MMS-TTS [25]	synthetic	TTS, 5 Indian languages	800

**Fig. 4.** Genuine and synthetic training clips per language, after excluding languages that have no genuine speech. Every language used for training has both classes.

Common Voice [26], is future work; because Common Voice audio is MP3-compressed and entirely genuine, it would need the synthetic clips in the same languages to receive the same MP3 round trip, or MP3 artefacts would become a label cue.

LibriSeVoc sampling. LibriSeVoc provides each of its 13,201 LibriSpeech utterances re-synthesised by all six vocoders. Taking the first 4,200 files per vocoder by name would give every vocoder the same utterances from the lowest-numbered speakers and leave two-thirds of speakers genuine only. We instead shuffle the utterances with a fixed seed and give each vocoder a different slice, so 11,999 utterances have two synthetic versions, 1,202 have one, and every genuine utterance has at least one synthetic counterpart.

7.3 Channel Conditioning

Every training clip, genuine or synthetic, receives one additional conditioned copy drawn from a single seeded sampler (Table 6): one acoustic effect followed by one channel codec. The acoustic effect models the room and the device; the codec models the network. The seed for each clip is a hash of its file name, so the draw is identical on every machine.

Applying the same sampler to both classes and every corpus is deliberate. Our earlier codec, RawBoost and reverberation copies were applied to ASVspoof only, so

Table 6. Channel-conditioning sampler. Each clip gets one acoustic effect and one codec, each drawn uniformly.

Stage	Options
Acoustic	room impulse response (RIRS_NOISES [27], 60,325 responses) · MUSAN [28] noise at 0–15 dB SNR, music at 5–15 dB, or 3–7-speaker babble at 13–20 dB · pink noise at 5–20 dB · RawBoost [14] (linear and non-linear convolutive plus impulsive noise)
Codec	none · G.711 μ -law at 8 kHz [29] (PSTN and contact centres) · AMR-WB at 6.6, 12.65 or 23.85 kbit/s [30] (VoLTE) · Opus at 8, 12, 16 or 24 kbit/s [31] (WebRTC, Meet, VoIP apps)

the model could learn “augmented” as a marker of that corpus and, since ASVspoof is 90% synthetic, of the synthetic class. The trainer checks that the share of conditioned rows matches between genuine and synthetic speech, overall and within each language, to within five percentage points, and stops if it does not.

7.4 Objective

Each training clip i belongs to a group $g(i) = (\text{language}, \text{generator}, \text{label})$, where the generator is the vocoder, attack system or corpus family and is “bona fide” for genuine speech. With N clips in G groups and n_g clips in group g , clip i receives weight

$$w_i = \frac{N}{G n_{g(i)}}, \quad (2)$$

so every group contributes equally regardless of size, and the mean weight is one. Earlier heads used a single positive-class weight, which balances genuine against synthetic overall but leaves the language prior intact: in the v3 training set, 89.8% of English clips were synthetic against 48.7% of Indian clips. Because the label is part of the group key, group weighting removes that prior as well.

The head has a shared trunk (Linear-ReLU-Dropout) and two outputs: a spoof logit f and a generator-family classifier h . It is trained on

$$\mathcal{L} = \frac{\sum_i w_i \ell_{\text{BCE}}(f(\mathbf{x}_i), y_i)}{\sum_i w_i} + \lambda \frac{\sum_i w_i \ell_{\text{CE}}(h(\mathbf{x}_i), z_i)}{\sum_i w_i}, \quad (3)$$

with y_i the label, z_i the generator family and $\lambda = 0.3$. Naming the generator is a language-independent task, so the auxiliary output discourages the trunk from answering with the language. A one-class objective (OC-Softmax [18], margins 0.9 and 0.2, scale 20) is available in place of binary cross-entropy for comparison.

7.5 Development Split and Model Selection

The development set is carved from training data by recording, and for IndicVoices and LibriSeVoc by speaker, so no speaker appears on both sides. Conditioned copies share

their source clip’s recording identifier and therefore land on the same side. Each group keeps at least one development recording.

For model selection, each development cell c is one (language, generator) spoof group scored against genuine speech of the same language; a cell falls back to all genuine speech only when its language has fewer than 20 genuine development clips, and such cells are reported as unmeasured. The checkpoint is chosen by

$$\theta^* = \arg \min_{\theta} \max_c \text{EER}_c(\theta), \quad (4)$$

so a language or generator cannot quietly fail while the average looks good. The mean cell EER is reported alongside, because worst-group selection can pull the mean down.

7.6 Training Details

Embeddings are standardised with training statistics only, which are stored in the checkpoint. The head has 256 hidden units and dropout 0.3, and is trained with Adam (learning rate 10^{-3} , batch 256) for up to 20 epochs with early stopping after 6 epochs without improvement in worst-group EER. The checkpoint records the full provenance of the run: every input root, the layer and pooling configuration, the group table with train and development counts, the per-cell development EER and the exact command line. The trainer also refuses to run if the held-out IndicVoices set appears among its inputs, if a shard’s embeddings contain non-finite values, if a folder’s labels contradict its content (a synthetic folder labelled genuine), if two inputs were padded differently, or if a language cannot be determined.

8 Data Engineering and Audit

8.1 Extraction

All audio is resampled to 16 kHz mono, cropped to its first 4.0 s and passed once through the frozen front-end. Embeddings are written in shards of 100 rows, each with a label file and a file-name list. Shards are written atomically and the embedding file last, so an interrupted run is resumed by re-running the same command and a half-written shard is never mistaken for a finished one; if a batch fails, its clips are retried one at a time, and a shard that still cannot be completed is not written. Extraction ran in parallel on several machines. To confirm that embeddings from different machines can be mixed, the same shard was extracted on two of them; the worst row-wise cosine similarity was 0.99997.

8.2 Silence Padding

The extractor originally zero-padded clips shorter than 4 s. Clip lengths differ sharply between corpora (Fig. 5): 71.1% of ASVspoof clips and 61.0% of In-the-Wild clips are shorter than 4 s, against 9.7% of MLAAD and 3.8% of IndicSynth. A silent tail therefore identified the corpus, and through it the likely label. It also differed from deployment, where a live window is always full. All training and test embeddings were converted to

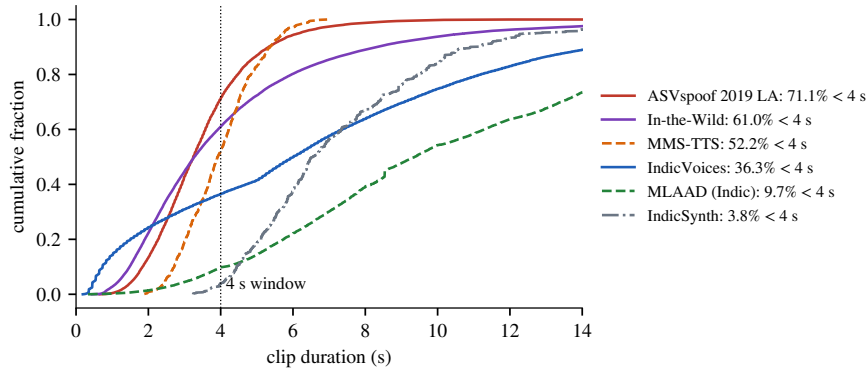


Fig. 5. Clip-duration distribution per corpus. Legend: share of clips shorter than the 4 s window.

repeat-padding, in which a short clip is tiled to fill the window; only the short clips had to be re-extracted, since longer clips are cropped identically under both schemes. For a 1 s genuine clip, the earlier head’s score changed from 0.8795 with zero-padding to 0.0329 with repeat-padding, while a full-length control was unchanged.

8.3 Speaker Overlap

The IndicVoices recording lists we started from were disjoint by recording but not by speaker: 682 of the 937 held-out speakers also had recordings in training. We report held-out results on the speaker-disjoint subset of 2,551 clips from 256 speakers (Kannada, Hindi, Punjabi, Gujarati, Tamil, Marathi, Bengali and Telugu), with the full held-out set as a secondary figure. Eleven recording sessions were split across both lists; their 173 training chunks were removed and kept in the held-out set.

8.4 Labels

Folders of audio are extracted with a placeholder label, so every synthetic folder is stamped with its label after extraction and checked; the trainer rejects any folder whose labels contradict its generator family.

9 Building the Final Model (v4)

Internally, the final model is SONIX v4, the fourth generation after the baseline, the robust heads and v3. This section records how it was built: the stages, every embedding root with its exact row count, the re-extraction, the conditioning runs, the group structure the trainer sees, the checks it must pass and the ablations planned around it.

Table 7. Datasets across SONIX generations. T = training, E = evaluation only, – = not used.

Dataset	Content	baseline / aug.	robust / v2	full_ho	v3	v4
ASVspoof 2019 LA train [5]	English, genuine + 6 attacks	T	T	T	T	T
ASVspoof 2019 LA eval [5]	71,237 clips, mostly unseen attacks	E	E	E	E	E
G.711 copies of ASVspoof	telephone codec, ASVspoof only	T (aug.)	T	T	T	– ^a
RawBoost copies [14]	channel/impulsive noise, ASVspoof only	–	T	T	T	– ^a
RIR + MUSAN copies [27,28]	reverb and noise, ASVspoof only	–	T (v2)	T	T	– ^a
IndicVoices [21]	genuine Indian speech	–	–	T	T	T
IndicVoices held out	5,050 clips; 2,551 speaker-disjoint	–	–	E	E	E
ASVspoof 5 [7]	crowdsourced, newer attacks	–	–	T ^b	–	E ^c
MMS-TTS [25]	Indian TTS, 5 languages	–	–	–	T	T
MMS-TTS augmented	channel copies, fakes only	–	–	–	T	– ^a
IndicSynth [22]	Indian voice conversion, 12 languages	–	–	–	T	T ^d
IndicSynth augmented	channel copies, fakes only	–	–	–	T	– ^a
MLAAD [23]	Indian TTS, 19 models	–	–	–	–	T ^d
LibriSeVoc [24]	English genuine + 6 vocoders	–	–	–	–	T
Symmetric _cond copies	all v4 training folders	–	–	–	–	T
In-the-Wild [8]	31,779 real-world clips	E	E	E	E	E
ASVspoof 2021 DF [6]	compressed deepfakes	E	E	E	E	E
Matched recordings	21 + 20 genuine/cloned	E	E	E	E	E
DEEP-VOICE [20]	retrieval-based VC	–	–	–	–	E

^aReplaced in v4 by one symmetric conditioned copy of every training folder, because one-sided copies became a corpus or class marker. ^bIn the full_indic_as5 head only. ^cPlanned. ^dWithout Malayalam, Urdu, Odia and Sanskrit.

9.1 Datasets and Techniques Across Versions

Table 7 lists every dataset SONIX has used from the first head to v4 and what it was used for in each generation, and Table 8 does the same for the techniques. Datasets marked E are used only for evaluation in that generation.

9.2 Build Stages

The build was planned as ten stages, each with an exit condition (Table 9). The order matters: the diagnosis came first because it decided that the representation, not only the data, had to change.

9.3 Embedding Roots

All clean and conditioned roots use the same configuration: XLS-R 300M, hidden states 5, 6 and 24, mean and standard-deviation pooling, 6,144 dimensions stored as 16-bit floats (about 12 KB per clip), shards of 100 rows, repeat-padding. Each root carries a meta.json describing its layers, pooling, dimensions and padding, and the trainer refuses roots whose descriptions disagree. Table 10 lists every root. All clean roots passed the automated root check (row counts, labels, padding, dimensions, finite values, intact shards), and the conditioned roots are checked the same way before they are copied for training.

After dropping Malayalam, Urdu, Odia and Sanskrit (2,000 MLAAD and 5,600 Indic-Synth clips), the clean training set has 129,640 rows, and 259,280 with the conditioned copies.

Table 8. Techniques across SONIX generations.

Component	Earlier heads (baseline to v3)	v4
Front-end	XLS-R 300M, frozen	XLS-R 300M, frozen
Representation	final-layer output, mean over time, layers 5, 6, 24, mean + std, 6,144-d 1,024-d	
Short clips	zero-padded to 4 s	repeat-padded to 4 s
Head	MLP 1024–256–1, dropout 0.3	shared trunk (256, dropout 0.3) + spoof output + generator-family output
Loss	BCE with one positive-class weight	group-weighted BCE (or OC-Softmax) + 0.3 × generator cross-entropy
Imbalance handling	genuine vs. synthetic only	language × generator × label groups
Augmentation	G.711, RawBoost, RIR, MUSAN on ASVspoof (and on Indian fakes in v3)	one shared sampler on every training folder: RIR / MUSAN / pink / RawBoost × none / G.711 / AMR-WB / Opus
Development split	random or by recording	by recording, by speaker for IndicVoices and LibriSeVoc
Model selection	mean development EER	worst-group development EER
Languages in training	English (+ Indian genuine in full_ho; + 12 Indian fake in v3)	8 Indian + English, genuine and synthetic in each
Safety checks	shape of embeddings	finite values, labels vs. folder, padding mode, channel symmetry, held-out guard, language required

9.4 Repeat-Padding Conversion

Roots first extracted with zero-padding were converted in place: only rows whose source audio is shorter than 4 s were re-embedded, and a root is marked as repeat-padded only when every shard is done (Table 11). The converted counts match the independently measured duration statistics exactly, which confirms that every short clip, and nothing else, was re-embedded. ASVspoof 2019 LA train was re-extracted from scratch with the new extractor instead.

9.5 Conditioning Runs

Twelve training folders were conditioned with one sampler (identifier 637f7a3c23eb, seed 0), drawing from 60,325 room impulse responses and 930 noise, 660 music and 426 speech recordings from MUSAN. Every run finished with no failed clips, and the verification step confirmed that all twelve used the same sampler configuration. Each conditioned copy is cropped to 4 s before processing, keeps its source file name, and is recorded with the exact effect, codec, bitrate and SNR drawn for it, so the per-codec and per-effect breakdowns in the results can be computed from the records.

Table 9. v4 build stages.

Stage	Name	Content and exit condition
S0	Diagnose	Probes A–C, layer sweep, matched-recording test of all earlier heads (Sect. 6, 11)
S1	Metadata	Language, generator family, recording and speaker for every clip; no recording in two splits
S2	Acquire	IndicVoices, IndicSynth, MLAAD, LibriSeVoc, In-the-Wild, MUSAN, RIRS_NOISES
S3	Condition	One conditioned copy per training clip from one shared sampler; symmetry verified
S4	Extract	Layers 5, 6, 24 with mean and standard deviation, 6,144 dimensions, repeat-padded
S5	Compose	Drop languages without genuine speech; seeded LibriSeVoc slices; group table
S6	Train	Group-balanced loss, generator-family auxiliary head, OC-Softmax ablation
S7	Select	Worst-group EER on a speaker-disjoint development set
S8	Evaluate	In-the-Wild, ASVspoof 2019 eval, 256 unseen Indian speakers, matched recordings, DEEP-VOICE
S9	Integrate	6,144-d front-end in the live service, per-domain thresholds, real-time factor

9.6 Group Structure

Group identity comes from the root name and the file name: the corpus and vocoder from the root, the language from the file name or, for IndicVoices, from the per-chunk metadata, and the recording or speaker identifier from the file name or metadata (Table 12). Languages are normalised to ISO codes, so that for example “bengali”, “ben” and “bn” become one group. The result is 39 training groups: 9 genuine groups (8 Indian languages and English) and 30 synthetic groups (IndicSynth in 8 languages, MLAAD in 5, MMS-TTS in 5, and English with 6 ASVspoof attacks and 6 vocoders).

9.7 Training Run and Ablations

The model is trained on 12 clean and 13 conditioned training roots with group derivation from names and metadata, the IndicVoices chunk table for language and speaker, and the four languages above dropped from the IndicSynth and MLAAD roots and their conditioned copies. Around the main run we compare (Table 13): binary cross-entropy against OC-Softmax; the auxiliary generator head on and off; with and without conditioned copies; and single-layer against three-layer representations. Each ablation changes one factor and is selected with the same worst-group rule.

9.8 Acceptance Gate

v4 becomes the service default only if it passes all of the following automated checks.

1. The IndicVoices held-out root is absent from the training inputs.
2. No recording or speaker appears in both training and development data.

Table 10. v4 embedding roots. Every training root also has a conditioned copy (`_cond`) with the same row count.

Root	Content	Rows	Label Use
<code>embeddings_v2/train</code>	ASVspoof 2019 LA train	25,380	2,580 / 22,800 train
<code>_mlaad</code>	MLAAD Indian TTS	19,000	1 train
<code>_indicsynth</code>	IndicSynth	16,800	1 train
<code>_indicvoices_tr</code>	IndicVoices train	36,859	0 train
<code>_librisevoc_gt</code>	LibriSeVoc genuine	13,201	0 train
<code>_librisevoc_<vocoder></code>	6 vocoders	$6 \times 4,200$	1 train
<code>_mms_tts</code>	MMS-TTS	800	1 train
<code>embeddings_v2/eval</code>	ASVspoof 2019 LA eval	71,237	protocol test
<code>embeddings_v2/itw</code>	In-the-Wild	31,779	19,963 / 11,816 test
<code>_indicvoices_ho</code>	IndicVoices held out	5,050	0 test

Table 11. Rows re-embedded when converting from zero- to repeat-padding.

Root	Short rows	Total	Share
ASVspoof 2019 LA eval	55,388	71,232*	77.8%
In-the-Wild	19,385	31,779	61.0%
IndicVoices train	13,324	36,859	36.1%
IndicVoices held out	1,901	5,050	37.6%
MLAAD	1,846	19,000	9.7%
IndicSynth	640	16,800	3.8%
MMS-TTS	418	800	52.3%
LibriSeVoc (7 roots)	1	38,401	<0.01%

*One shard had lost 5 rows to an earlier extractor fault; it was re-extracted, giving 71,237.

3. Conditioned rows are shared alike by genuine and synthetic speech, overall and per language.
4. No training group is too small to estimate reliably, or it is merged into a coarser group.
5. Probe A (language) accuracy is lower than 99.33% on the new representation.
6. Probe B (corpus) accuracy is lower than 100%.
7. Probe C cross-language AUC is higher than 0.5196.
8. Scores of genuine held-out speech do not rise or fall with its signal-to-noise ratio.
9. The padding test (1, 2, 3 and 66 s clips) reproduces the recorded values.
10. AUC on the Indian subset of the matched recordings exceeds 0.90.
11. In-the-Wild EER is reported and falls within the range of published open-set results.
12. Degradation on DEEP-VOICE, a generator family absent from training, is measured.
13. Scoring keeps up with the 0.5 s hop in real time.

Checks 5–7 are the ones that show the shortcuts found in Sect. 6 are gone, and they are measurements the reference literature does not report.

Table 12. How each root’s clips are assigned to groups.

Root	Language	Generator	Recording / speaker unit
ASVspoof 2019 LA	English	attack id from protocol (A01–A06, or bona fide)	utterance
IndicVoices	chunk metadata	bona fide	speaker (from metadata)
IndicSynth	file name	IndicSynth	clip
MLAAD	file name	MLAAD	clip
MMS-TTS	file name	MMS-TTS	clip
LibriSeVoc	English	vocoder from root name, or bona fide	LibriSpeech speaker

Table 13. Planned ablations around the main v4 run.

Ablation	Question it answers
BCE vs. OC-Softmax	Does a one-class objective help on unseen generators (DEEP-VOICE, In-the-Wild)?
$\lambda = 0.3$ vs. $\lambda = 0$	Does naming the generator reduce reliance on language?
With vs. without _cond roots	How much does symmetric channel conditioning buy on real calls?
Layers 5,6,24 vs. 24 only vs. 6 only	Is the multi-layer representation needed?
Group weighting vs. class weighting	Does group balancing remove the language prior in practice?

9.9 Integration

The live front-end still produces the 1,024-dimensional representation used by the earlier heads. For v4 it will compute the same three-layer, 6,144-dimensional vector as extraction, and each head in the registry declares which representation it expects, so a head can never be scored on the wrong input. The earlier heads stay loaded for side-by-side comparison in the console. Band thresholds are then fitted per domain on held-out data and stated as domain-specific, and the real-time factor is re-measured.

9.10 Timeline

Table 14 summarises the timeline.

10 Evaluation Protocol

None of the evaluation sets in Table 15 is used for training, augmentation or threshold fitting of the model being evaluated.

Table 14. v4 timeline (2026).

Date	Milestone
19 Sep	Probes A–C and INLP run; layer sweep; matched-recording test of all earlier heads
20 Sep	v4 build plan written; three-layer extractor written
21–24 Sep	Datasets acquired; IndicVoices re-fetched by chunk list and its train/held-out overlap fixed
24–25 Sep	All clean roots extracted; group-aware trainer written
25 Sep	Silence-padding bias found and fixed; all roots converted; twelve folders conditioned and extracted

Table 15. Evaluation sets.

Set	Clips Purpose
In-the-Wild [8]	31,779 Real-world generalisation (19,963 genuine, 11,816 deepfake)
ASVspoof 2019 LA eval [5]	71,237 Standard benchmark; mostly unseen attacks
IndicVoices held out [21]	2,551 False alarms on 256 unseen Indian speakers (full set: 5,050)
Matched recordings	21 / 20 Genuine and cloned speech of the same speakers (two sets)
DEEP-VOICE [20]	– Retrieval-based voice conversion, a generator family absent from training
ASVspoof 2021 DF [6]	– Compressed deepfake track (reported on 75% key coverage)

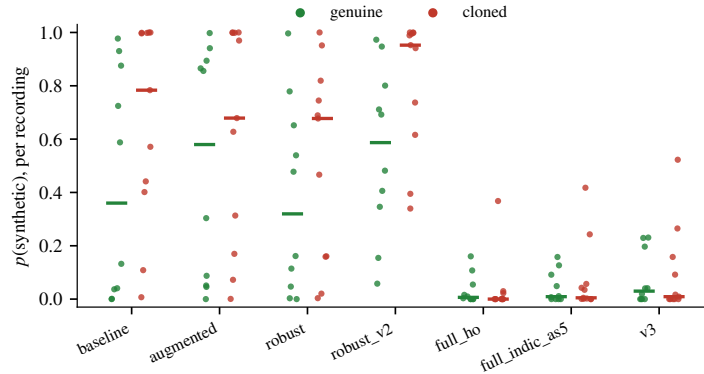
Metrics. We report equal error rate (EER) and AUC, never accuracy: the benchmark data is about 90% synthetic, so accuracy rewards a detector that always says “fake”. On the Indian held-out set, which is all genuine, we report the false-alarm rate per 4 s window, never per call. Scores are $p(\text{synthetic})$, so higher means more likely fake.

Matched recordings. The most direct test is a set of genuine and cloned recordings of the same speakers. The first set has 21 recordings (10 genuine, 11 cloned) of public figures with 9 true genuine–clone pairs; the second has 20 recordings (10 and 10) identified only by number. A recording’s score is the mean over its windows. With sets this small the ordering of heads is more reliable than any single value. Recordings of public figures are used for internal evaluation only and are not published; a volunteer set of more than 200 pairs, recorded with consent across phones, laptops and calls, is being collected to replace them.

Acceptance checks. Before a model can become the service default it must pass automated checks: the held-out set is absent from training; no recording appears in two splits; conditioning is symmetric; re-running probes A, B and C on the new representation shows lower language and corpus decodability and higher cross-language AUC; the score does not track the signal-to-noise ratio of genuine speech; the padding test reproduces the recorded values; and In-the-Wild EER falls in the range published systems achieve.

Table 16. Earlier heads on matched genuine and cloned recordings. Named set: $n = 21$, 9 true pairs. Numbered set: $n = 20$. Medians are recording-level $p(\text{synthetic})$.

Head	Named set ($n = 21$)			Numbered set ($n = 20$)	
	AUC	median genuine	median cloned	AUC	median genuine
robust_v2	0.764	0.587	0.952	0.820	0.758
baseline	0.709	0.360	0.784	0.760	0.521
augmented	0.655	0.580	0.679	0.730	0.648
robust	0.627	0.320	0.677	0.750	0.633
full_indic_as5	0.536	0.009	0.005	0.720	0.000
v3	0.509	0.030	0.009	0.640	0.002
full_ho	0.355	0.006	0.000	0.290	0.000

**Fig. 6.** Recording-level scores of each earlier head on the named matched set ($n = 21$, 9 true pairs). Bars mark medians. The heads trained with Indian speech score almost everything as genuine.

11 Results

11.1 Earlier Heads on Matched Recordings

Table 16 and Fig. 6 compare the earlier heads on the matched recordings. The `_rebuild` checkpoints are re-serialised copies of `baseline` and `augmented` and score identically, so they are omitted.

Three patterns stand out. The best head, `robust_v2`, never saw any Indian speech; its advantage comes from room reverberation and noise, which put room tone rather than digital silence into its genuine training audio. The three heads trained with IndicVoices assign near-zero scores to almost every recording, genuine or cloned, so they rank at or below chance: they learned that Indian speech is genuine. And even the best head gives genuine recordings a median score of 0.59, so on real call audio it would raise many false alarms. This is the behaviour the diagnosis predicts, and the final model is built to remove it.

Table 17. Published EER (%) on public evaluation sets, all trained on ASVspoof 2019 LA.

System	Front-end	ASVspoof19 eval	In-the-Wild
RawNet2 [8]	raw waveform	3.15	37.82
RawGAT-ST [8]	raw waveform	1.23	37.15
XLS-R + SLS [13]	XLS-R 300M, fine-tuned	–	7.46

11.2 Comparison with Published Systems

Table 17 lists published results on the public evaluation sets SONIX uses. They set the reference points: systems trained on ASVspoof 2019 LA alone reach about 37% EER on In-the-Wild, and the strongest fine-tuned XLS-R system reaches 7.46%. Unlike that system, SONIX keeps its front-end frozen, which is what lets it serve several heads from one front-end in real time.

12 Discussion

The central finding is that a detector can look excellent on its own benchmark while measuring the wrong thing. The earlier heads were not badly trained; they learned what the data allowed. When genuine Indian speech came from one corpus and synthetic speech from others, “Indian” and “corpus” were the easiest features to use, and the representation offered them almost perfectly. Adding more data of the same shape made this worse, not better: `full_ho` had the lowest false-alarm rate on Indian speech of any head and the worst ranking on matched recordings.

The same lesson applies to augmentation. An effect applied to one corpus or one class becomes a feature of that corpus or class. We therefore treat symmetry as a property to check, not a convention to follow: the trainer measures it and refuses asymmetric data. Silence padding is the same problem in its simplest form.

For deployment, the swappable head matters more than any single score. New cloning tools appear faster than a front-end can be retrained; with a frozen front-end and cached embeddings, a new head can be trained in minutes and served from the same endpoint without redeployment.

13 Limitations

- *Matched test sets are small.* With 21 and 20 recordings, head orderings are informative but individual AUCs are not precise. The volunteer set addresses this.
- *Language coverage.* The final model trains on 8 Indian languages plus English. Behaviour on other languages, including those excluded for lack of genuine speech, is not established.
- *Generators.* Training spoofs cover TTS, voice conversion, neural vocoders and ASVspoof attacks, but not current zero-shot cloning systems such as XTTS [32] or OpenVoice [33].

- *Mixed audio*. A browser tab carries all remote participants in one stream, so a score applies to a window, not to a named speaker; SONIX does not perform diarisation.
- *Client audio processing*. Meet and similar clients apply noise suppression and gain control before SONIX receives the audio, which adds domain shift the training data only partly covers.
- *Latency*. The first verdict needs one full 4 s window.
- *Platforms*. The extension supports Chrome and Edge only, and ordinary mobile calls are out of scope.

14 Ethics, Privacy and Licensing

Monitoring a call requires the knowledge of the people on it, and recording law differs by jurisdiction. SONIX enforces operator consent on the server, stores no audio and keeps an audit record of scores and events only. It flags; people decide, and every flag is a prompt to verify through another channel, not an accusation. Recordings of public figures used in evaluation are kept internal and are not redistributed. MLAAD is licensed CC BY-NC 4.0, which permits research and this prototype but not commercial use; a commercial deployment would retrain without it. IndicVoices is licensed CC BY 4.0.

15 Future Work

The next steps are: zero-shot cloned speech from XTTS [32] and OpenVoice [33] in Indian languages, with reference speakers kept disjoint from every test set; genuine speech for the excluded languages with MP3-matched synthetic counterparts; frame-level back-ends such as AASIST [11] and RawNet2 [12] as baselines on the same Indian test set; per-domain threshold calibration; the volunteer matched set; and a pilot with a contact-centre partner.

16 Conclusion

SONIX is a working real-time layer that flags cloned voices during calls without blocking them. Building it showed that the standard recipe of a frozen SSL embedding with a small head learns language and corpus rather than synthesis, and that adding Indian speech without care makes this worse. The final detector is designed around that measurement, and its evaluation is designed so that the same shortcuts cannot produce a good-looking number.

Team Contributions. Yugal Mahajan led the machine-learning work: extraction, training, probes, model design and evaluation. Navya acquired IndicVoices, MLAAD, MUSA-SAN and RIRS_NOISES and ran most of the final extraction and conditioning. Akshat prepared IndicSynth and LibriSeVoc and generated demonstration audio. Anubhav built the matched recording set and live capture, and ran codec evaluation. Suryansh built the operator console and the Chrome extension. Yukti built the metrics and calibration tooling and led the presentation.

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